

PF Lab 6 tasks(4)

Name: Hassan Ali Khan

Section: BCS-1K

Roll No: 25K-0516

Task 1

```
#include<stdio.h>

int main()
{
    int num,even=0,odd=0;

    char ch;

    printf("Enter a number\n");
    scanf("%d",&num);

    do{

        for(int i=1;i<=num;i++){
            scanf("%d",&num);

            if(num%2==0){

                printf("Number is even\n");

                even+=1;

            }

            else{

                printf("Number is odd\n");

                odd+=1;

            }

        }

        printf("Do you want to continue?\n");

        scanf(" %c", &ch);

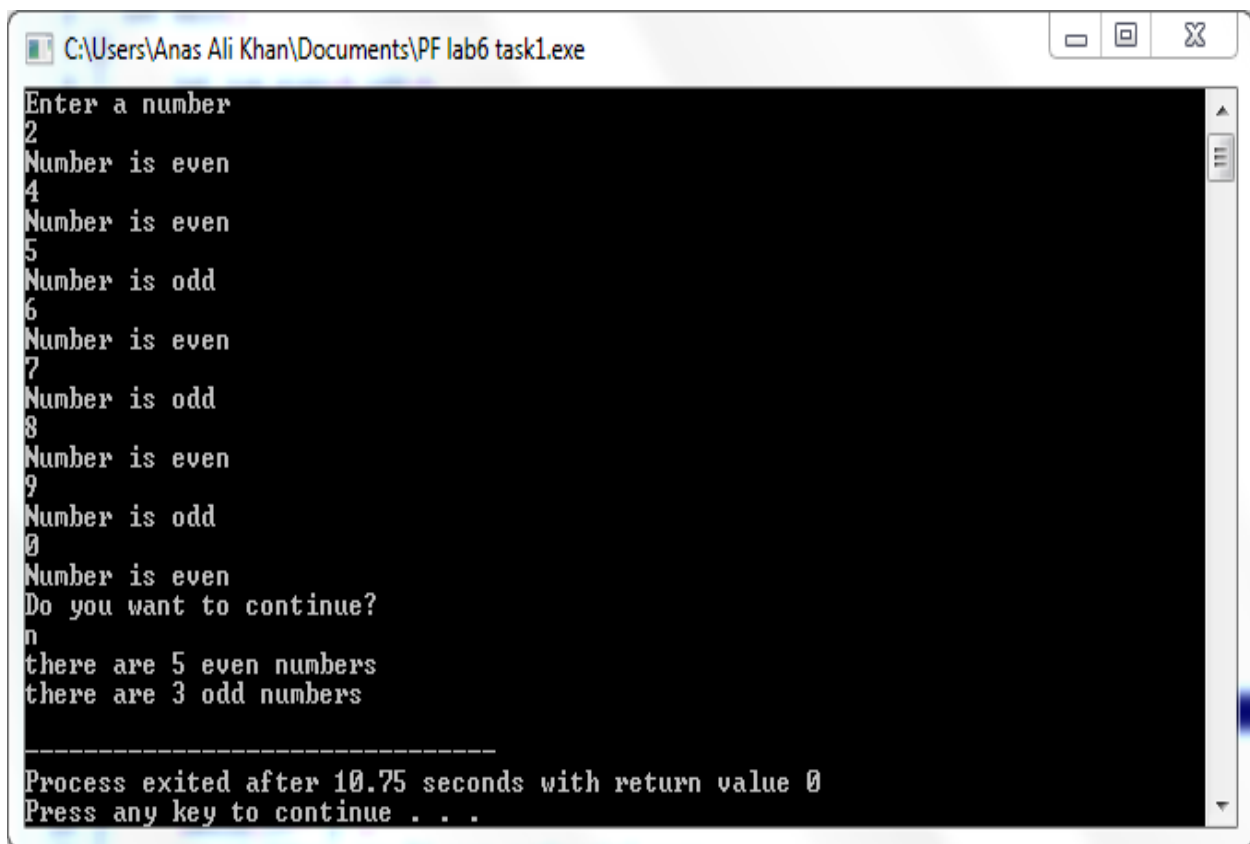
    }while(ch=='y');
```

```
printf("there are %d even numbers\n",even);
```

```
printf("there are %d odd numbers\n",odd);
```

```
return 0;
```

```
}
```



```
C:\Users\Anas Ali Khan\Documents\PF lab6 task1.exe
Enter a number
2
Number is even
4
Number is even
5
Number is odd
6
Number is even
7
Number is odd
8
Number is even
9
Number is odd
0
Number is even
Do you want to continue?
n
there are 5 even numbers
there are 3 odd numbers

-----
Process exited after 10.75 seconds with return value 0
Press any key to continue . . .
```

Task 2

```
#include <stdio.h>

int main()
{
    int num, i, j;

    int count = 0;

    int sum = 0;

    printf("Entee a number: ");

    scanf("%d",&num);

    printf("Prime numbers: ");

    for(i = 2; i <= num; i++) {

        for(j = 2; j < i; j++) {

            if(i % j == 0) {

                break;

            }

        }

        if(j == i) {

            printf("%d ", i);

            count++;

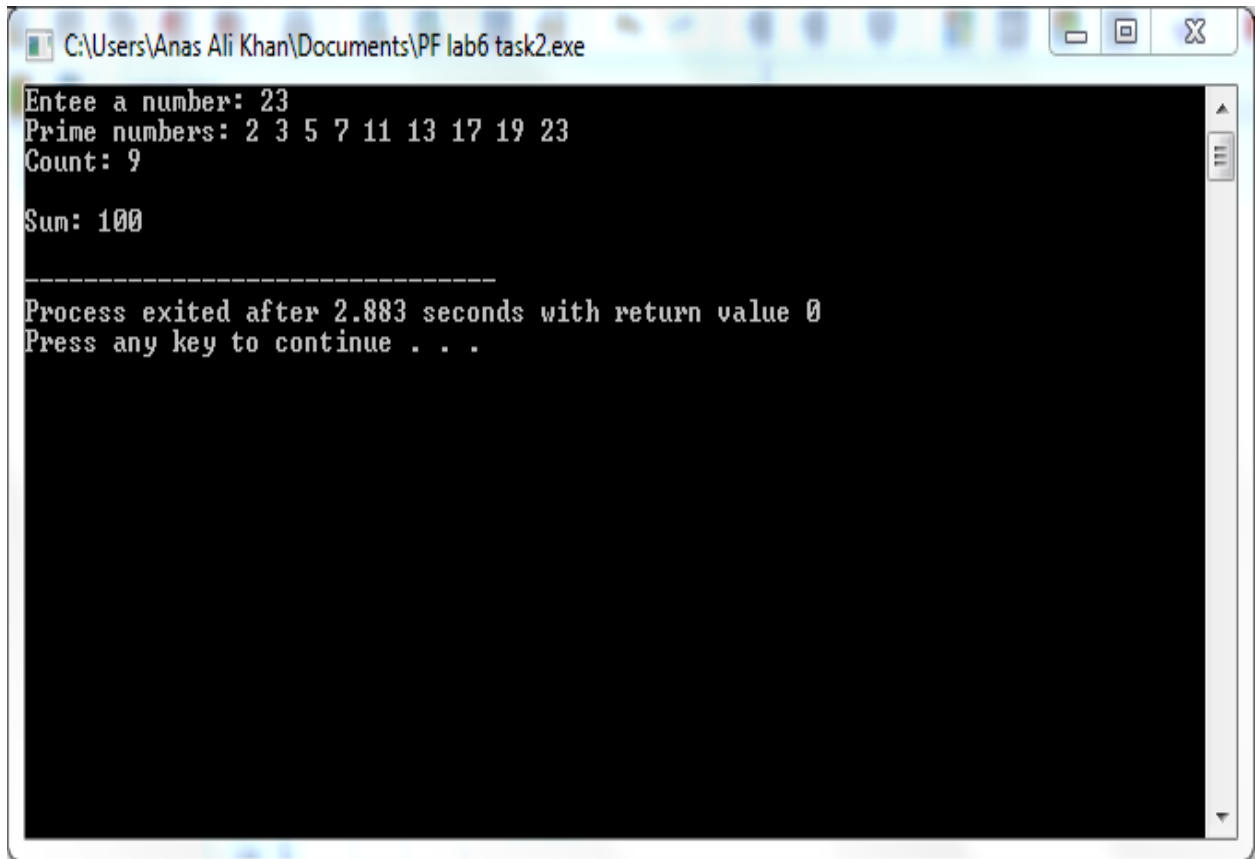
            sum += i;

        }

    }

}
```

```
    }  
}  
  
printf("\nCount: %d\n", count);  
  
printf("\nSum: %d\n", sum);  
  
return 0;  
}
```



```
C:\Users\Anas Ali Khan\Documents\PF lab6 task2.exe  
Entee a number: 23  
Prime numbers: 2 3 5 7 11 13 17 19 23  
Count: 9  
Sum: 100  
-----  
Process exited after 2.883 seconds with return value 0  
Press any key to continue . . .
```

Task 3

```
#include <stdio.h>

int main()
{
    int num;

    long first = 0, second = 1, next;

    long sum = 0;

    printf("Enter the number of terms: ");

    scanf("%d", &num);

    if(num <= 0) {
        printf("Please enter a positive number.\n");
    }

    printf("Fibonacci sequence: \n");

    for(int i = 1; i <= num; i++) {
        printf("%ld \n", first);

        sum = sum + first;

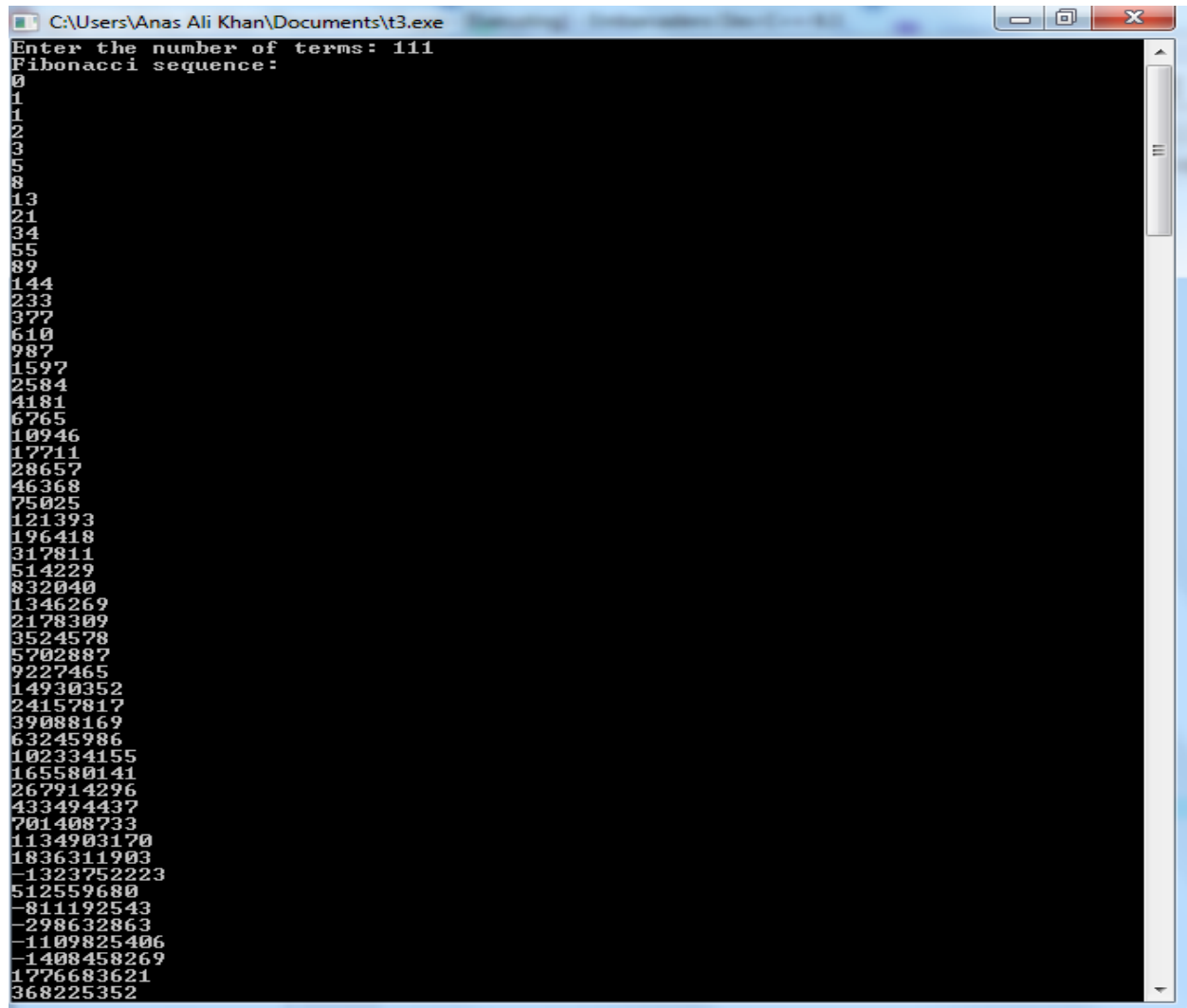
        next = first + second;
        first = second;
        second = next;
    }
}
```

```
}
```

```
printf("\nSum of Fibonacci sequence: %lld\n", sum);
```

```
return 0;
```

```
}
```



The screenshot shows a Windows command prompt window titled "C:\Users\Anas Ali Khan\Documents\t3.exe". The prompt asks "Enter the number of terms: 111" and "Fibonacci sequence:". The output displays the first 111 terms of the Fibonacci sequence, starting with 0 and 1, and ending with 368225352. The sequence is printed on a black background with white text.

```
C:\Users\Anas Ali Khan\Documents\t3.exe
Enter the number of terms: 111
Fibonacci sequence:
0
1
1
2
3
5
8
13
21
34
55
89
144
233
377
610
987
1597
2584
4181
6765
10946
17711
28657
46368
75025
121393
196418
317811
514229
832040
1346269
2178309
3524578
5702887
9227465
14930352
24157817
39088169
63245986
102334155
165580141
267914296
433494437
701408733
1134903170
1836311903
-1323752223
512559680
-811192543
-298632863
-1109825406
-1408458269
1776683621
368225352
```

```
C:\Users\Anas Ali Khan\Documents\t3.exe
695895453
-1958435240
-1262539787
1073992269
-188547518
885444751
696897233
1582341984
-2015728079
-433386095
1845853122
1412467027
-1036647147
375819880
-660827267
-285007387
-945834654
-1230842041
2118290601
887448560
-1289228135
-401779575
-1691007710
-2092787285
511172301
-1581614984
-1070442683
1642909629
572466946
-2079590721
-1507123775
708252800
-798870975
-90618175
-889489150
-980107325
-1869596475
1445263496
-424332979
1020930517
596597538
1617528055
-2080841703
-463313648
1750811945
1287498297

Sum of Fibonacci sequence: 30841242

-----
Process exited after 3.223 seconds with return value 0
Press any key to continue . . .
```


Task 4

```
#include<stdio.h>

int main()
{
    char ch,dec;

    int v=0,c=0;

    printf("Enter a single character\n");

    do{

        scanf(" %c",&ch);

        if
(ch=='a' || ch=='A' || ch=='e' || ch=='E' || ch=='i' || ch=='I' || ch=='o' || ch=='O' || ch=='u' || ch=='U'){

            printf("%c is vowel\n",ch);

            v+=1;

        }

        else{

            printf("%c is consonent\n",ch);

            c+=1;

        }

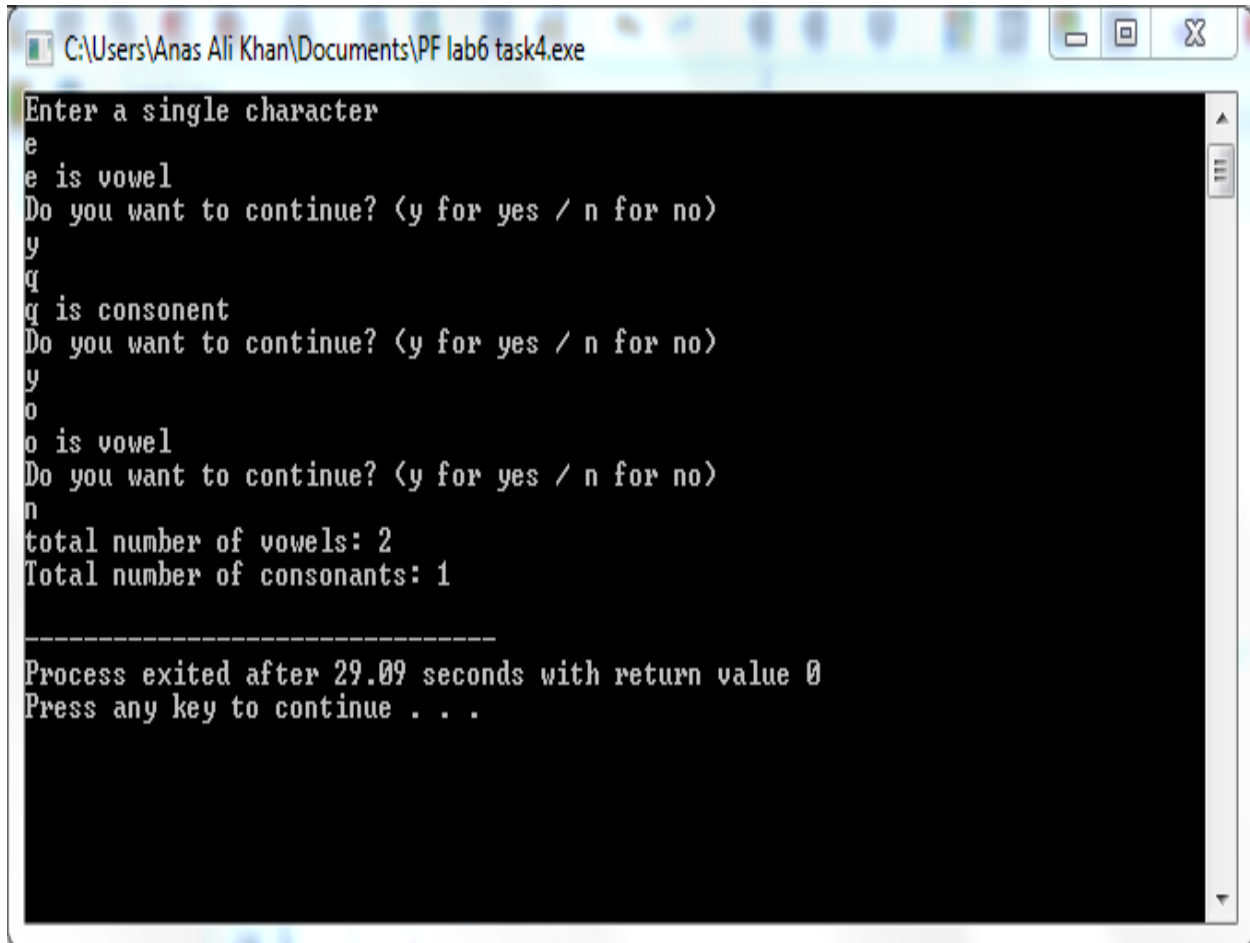
        printf("Do you want to continue? (y for yes / n for no)\n");

        scanf(" %c",&dec);

    }while(dec == 'y');

    printf("total number of vowels: %d\n",v);
```

```
printf("Total number of consonants: %d\n",c);  
  
return 0;  
  
}
```



```
C:\Users\Anas Ali Khan\Documents\PF lab6 task4.exe  
Enter a single character  
e  
e is vowel  
Do you want to continue? (y for yes / n for no)  
y  
q  
q is consonant  
Do you want to continue? (y for yes / n for no)  
y  
o  
o is vowel  
Do you want to continue? (y for yes / n for no)  
n  
total number of vowels: 2  
Total number of consonants: 1  
  
-----  
Process exited after 29.09 seconds with return value 0  
Press any key to continue . . .
```